Expasy

ProtParam (https://web.expasy.org/protparam)

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ProtParam - Results

User-provided sequence:

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        10         20         30         40         50         60
MKWVTFISLL FLFSSAYSRG VFRRDAHKSE VAHRFKDLGE ENFKALVLIA FAQYLQQCPF

        70         80         90        100        110        120
EDHVKLVNEV TEFAKTCVAD ESAENCDKSL HTLFGDKLCT VATLRETYGE MADCCAKQEP

       130        140        150        160        170        180
ERNECFLQHK DDNPNLPRLV RPEVDVMCTA FHDNEETFLK KYLYEIARRH PYFYAPELLF

       190        200        210        220        230        240
FAKRYKAAFT ECCQAADKAA CLLPKLDELR DEGKASSAKQ RLKCASLQKF GERAFKAWAV

       250        260        270        280        290        300
ARLSQRFPKA EFAEVSKLVT DLTKVHTECC HGDLLECADD RADLAKYICE NQDSISSKLK

       310        320        330        340        350        360
ECCEKPLLEK SHCIAEVEND EMPADLPSLA ADFVESKDVC KNYAEAKDVF LGMFLYEYAR

       370        380        390        400        410        420
RHPDYSVVLL LRLAKTYETT LEKCCAAADP HECYAKVFDE FKPLVEEPQN LIKQNCELFE

       430        440        450        460        470        480
QLGEYKFQNA LLVRYTKKVP QVSTPTLVEV SRNLGKVGSK CCKHPEAKRM PCAEDYLSVV

       490        500        510        520        530        540
LNQLCVLHEK TPVSDRVTKC CTESLVNRRP CFSALEVDET YVPKEFNAET FTFHADICTL

       550        560        570        580        590        600
SEKERQIKKQ TALVELVKHK PKATKEQLKA VMDDFAAFVE KCCKADDKET CFAEEGKKLV


AASQAALGL
```

[Documentation (/protparam/protparam-doc.html) / Reference (/protparam/protpar-ref.html)]

Number of amino acids: 609
Theoretical pI: 5.92
Molecular weight: 69366.68

Amino acid composition: CSV format

Ala	(A)	63	10.3%
Arg	(R)	27	4.4%
Asn	(N)	17	2.8%
Asp	(D)	36	5.9%
Cys	(C)	35	5.7%
Gln	(Q)	20	3.3%
Glu	(E)	62	10.2%
Gly	(G)	13	2.1%
His	(H)	16	2.6%
Ile	(I)	9	1.5%
Leu	(L)	64	10.5%
Lys	(K)	60	9.9%
Met	(M)	7	1.1%
Phe	(F)	35	5.7%
Pro	(P)	24	3.9%
Ser	(S)	28	4.6%
Thr	(T)	29	4.8%
Trp	(W)	2	0.3%
Tyr	(Y)	19	3.1%
Val	(V)	43	7.1%
Pyl	(O)	0	0.0%
Sec	(U)	0	0.0%
	(B)	0	0.0%
	(Z)	0	0.0%
	(X)	0	0.0%

Total number of negatively charged residues (Asp + Glu): 98
Total number of positively charged residues (Arg + Lys): 87

Atomic composition:

Carbon	C	3076
Hydrogen	H	4833
Nitrogen	N	821
Oxygen	O	919
Sulfur	S	42

Formula: $C_{3076}H_{4833}N_{821}O_{919}S_{42}$
Total number of atoms: 9691

Extinction coefficients:
Extinction coefficients are in units of $M^{-1}\ cm^{-1}$, at 280 nm measured in water.
Ext. coefficient 41435
Abs 0.1% (=1 g/l) 0.597, assuming all pairs of Cys residues form cystines

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Ext. coefficient    39310
Abs 0.1% (=1 g/l)   0.567, assuming all Cys residues are reduced
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Estimated half-life:

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The N-terminal of the sequence considered is M (Met).
The estimated half-life is: 30 hours (mammalian reticulocytes, in vitro).
                            >20 hours (yeast, in vivo).
                            >10 hours (Escherichia coli, in vivo).
```

Instability index:

The instability index (II) is computed to be 39.07
This classifies the protein as stable.

Aliphatic index: 77.57
Grand average of hydropathicity (GRAVY):-0.354